RiverDOCNet 密西西比河流域溶解性有机碳时空重构

RiverDOCNet DOC DOC 357 22,571 MAE 2.1845 mg C/L 3 6 12 MAE 2.1942 2.2168 2.2866 mg C/L MAE 1.9313 mg C/L R^2 0.4587 DOC

1

dissolved organic carbon DOC DOC [1]

DOC [2]

— DOC

RiverDOCNet — DOC

2

2.1

DOC Lu DOC [3] Casas-Ruiz DOC [1]

2.2

Wampler DOC [4] DOC

2.3

DOC INCA-C [5] [6]

GRU [7] Transformer [8]
DOC DOC

3 RiverDOCNet

3.1

12
DOC DOC

$$\hat{y}_{i,t} = f_\theta(e_i, w_{i,t-11:t},\ G)$$

15
8
12

3.2

GELU 24 12 12
GRU 24 GRU
DOC

$$u_i = W_2\,GELU(W_1 e_i + b_1) + b_2$$

$$g_{i,t} = GRU(w_{i,t-11:t}),\ h_{i,t}^{(0)} = W_f[u_i \,\|\, g_{i,t} \,\|\, s_t] + b_f$$

3.3

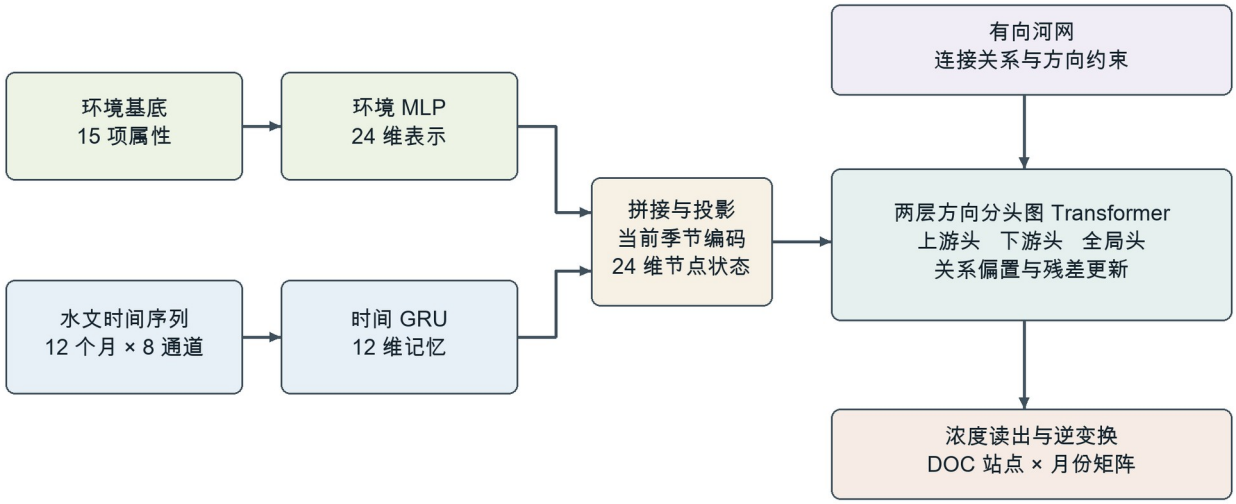
Transformer 8

DOC

$$\alpha_{ij,t}^{(h)} = softmax_{j \in N_i^{(h)}}\left(\frac{q_{i,t}^{(h)} \cdot (k_{j,t}^{(h)})^T}{\sqrt{d_h}} + b_h(r_{ij})\right)$$

DOC
log1p DOC 15
12,565

$$\hat{y}_{i,t} = max\{0,\ \exp[min(15,\ \mu + \sigma z_{i,t})] - 1\}$$



1 RiverDOCNet DOC

3.4

log1p DOC DOC

AdamW 0.001 0.0001 32 1 60 15 MAE 10

$$L = \frac{1}{|\Omega_{train}|} \Sigma_{(i,t)\in\Omega_{train}} \left[\frac{\log(1+y_{i,t}) - \mu}{\sigma} - z_{i,t} \right]^2$$

“RiverDOCNet ”

“RiverDOCNet ” GRU

4

4.1

357 324 1972 4 2026 9 654 DOC

USGS — NHDPlus NLDI StreamCat [9] 22,571 357×654

9.67% DOC

1

	104 7,262
	266 3 6 12
	232 54 71

DOC

DOC 565 569 579
DOC

4.2

DOC

MAE

4.3

MAE log1p MAE RMSE R^2
MAE RMSE mg C/L log-MAE R^2 R^2 1−SSE/SST R^2 R^2

$$MAE = \frac{1}{n}\Sigma_{k=1,\dots,n}|\hat{y}_k - y_k|, R^2 = 1 - \frac{\Sigma_k(y_k - \hat{y}_k)^2}{\Sigma_k(y_k - \bar{y})^2}$$

5

5.1

2

	M E↓	**log-M E↓**	**RMSE↓**	**R²↑**
RiverDOCNet	2.1845	0.3112	4.6436	0.5090
RiverDOCNet	**2.1839**	**0.3085**	4.6991	0.4972
	2.2526	0.3245	4.6955	0.4980
	2.3265	0.3578	4.6762	0.5022
	2.4512	0.4096	**4.5142**	**0.5361**
	2.4637	0.3560	4.8091	0.4735
	2.9912	0.3671	7.8566	−0.4067
	3.1652	0.4478	6.3018	0.0851
	3.0079	0.4796	5.1318	0.4004
	3.0433	0.4761	5.5570	0.2969
	3.0454	0.4838	5.2819	0.3648

	M E↓	log-M E↓	RMSE↓	R²↑
	3.1120	0.4832	5.7340	0.2515
	3.1275	0.4922	5.4553	0.3224
	3.5574	0.5475	6.8609	−0.0717
	3.6069	0.5568	6.0572	0.1647

MAE log-MAE 2.1839 0.3085 2.1845 0.3112
log-MAE RMSE 4.6436 R² 0.5090 RMSE 4.5142 R² 0.5361 RiverDOCNet

MAE —— —— MAE 27.40%
3.05% 6.13% 99.60% 99.92%

5.2

3 M E

	3↓	6↓	12↓
RiverDOCNet	**2.1942**	**2.2168**	**2.2866**
RiverDOCNet	2.2807	2.3083	2.3329
	2.2813	2.3101	2.2876
	2.3708	2.2876	2.3180
	2.2792	2.3048	2.3502
	2.2917	2.4632	2.6166
	2.3640	2.5419	2.7798
	2.4021	2.3852	2.3633
	2.4095	2.5648	2.6145
	2.3359	2.5882	2.7820

MAE 2.1942 2.2168 2.2866 3.79% 3.96% 1.98%
—

MAE 2.1830
12 MAE 2.2876 2.2866

5.3

4

	M E↓	log-M E↓	RMSE↓	R²↑	M E
	2.1839	**0.3085**	**4.6991**	**0.4972**	+0.00%
	3.6618	0.5640	6.6775	−0.0189	+67.68%
	2.2911	0.3380	4.7802	0.4792	+4.91%
	2.2137	0.3176	4.7402	0.4883	+1.36%

MAE 67.68% 4.91% 1.36%
MAE 2.20% MAE 0.04%
DOC
“ ” MAE 2.1749 2.1848 2.2028
GRU 2.2083 2.2254 GRU 0.16%
2 MLP GRU

5.4

5

	M E↓	**log-M E↓**	**RMSE↓**	**R²↑**
	1.9313	**0.2760**	**4.0633**	**0.4587**
	1.9646	0.2823	4.1257	0.4425
	2.0819	0.3022	4.2521	0.4079
	2.0593	0.2984	4.2237	0.4168

MAE 1.9313 2.0819 2.0593 7.24% 6.22%
MAE 1.70%

2 2.1845 1.9313

6

MAE RMSE R²

DOC 95.6% DOC 32.6%
DOC

7

RiverDOCNet MLP 12 GRU Transformer DOC

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